

Honors Calculus III, Homework 6

Please upload solutions to **Questions 1(c), 2, 3 and Problem 2-7** by **8pm on Monday, May 08**. You can use any results you want from lectures, provided you state them clearly, but make sure to write complete proofs. The remaining problems will not be graded, and you should not hand them in, but you are still expected to learn how to do them, unless it says “optional”. As usual, for full credit you need to show and explain your work.

- For each of the following sets, determine with proof whether they are open, closed, compact or connected.
 - $B(0, 1) \subset \mathbf{R}^n$.
 - $\overline{B}(0, 1) \subset \mathbf{R}^n$.
 - (5pt) $\{(x, y) \in \mathbf{R}^2 : (x + y - 4)(x^2 + y^2 - 1) = 0\}$.
 - $\{x \in \mathbf{R}^n : \max_i \{|x_i|\} \leq 1\}$.
 - $\{(x, y) \in \mathbf{R}^2 : y = \sin(x)\}$.
- (5pt) Let $f : \mathbf{R}^n \rightarrow \mathbf{R}$ be a linear map. Prove that there exists $y \in \mathbf{R}^n$ such that $f(x) = \langle x, y \rangle$ for all $x \in \mathbf{R}^n$.
- (5pt) Let $U = \{(x, y) \in \mathbf{R}^2 : y \neq 0\}$. Prove from the definition that the function

$$f : U \rightarrow \mathbf{R}, f(x, y) = x/y$$

is differentiable at $(1, 1)$ with derivative

$$Df(1, 1)(h, k) = h - k.$$

- Problems from Spivak, *Calculus on manifolds*: 2-2, 2-7(5pt), 2-8, 2-17, 2-20.

Solutions to (1c). Since the product of two real numbers is zero if and only if at least one of the factors is zero, the given set X is equal to

$$\{(x, y) \in \mathbf{R}^2 : x + y - 4 = 0\} \cup \{(x, y) \in \mathbf{R}^2 : x^2 + y^2 - 1 = 0\}.$$

These sets are closed, since they are zero sets of polynomial (hence continuous) functions. So X is closed. Since it contains the straight line of equation $x + y - 4 = 0$, it is not bounded, hence it is not compact, by the Heine–Borel theorem. Since X is not empty and not equal to $\mathbf{R}^2$ (for example, it does not contain $(5, 0)$ but it contains $(1, 0)$) it is not open because it is closed. Finally, notice that if $x^2 + y^2 = 1$ then $|x| \leq 1$ and $|y| \leq 1$, hence $|x + y| \leq 2$ by the triangle inequality, and so $x + y - 3 < 0$. On the other hand, if $x + y - 4 = 0$ then $x + y - 3 = 1 > 0$. This means that if we define the open sets

$$U^+ = \{(x, y) \in \mathbf{R}^2 : x + y - 3 < 0\} \text{ and } U^- = \{(x, y) \in \mathbf{R}^2 : x + y - 3 > 0\}$$

then $X \subset U^+ \cup U^-$, and $X \cap U^+$ and $X \cap U^-$ are disjoint and nonempty. So X is not connected. (Geometrically, X is the disjoint union of a circle and a straight line, which can be separated by the line $x + y - 3 = 0$, causing X to be disconnected.)

Solutions to 2. Let $\{e_1, \dots, e_n\}$ be the standard basis of $\mathbf{R}^n$ and define

$$y = \sum_{i=1}^n f(e_i)e_i.$$

To show that $f(x) = \langle x, y \rangle$ for all $x \in \mathbf{R}^n$ it suffices to show this for $x = e_1, \dots, e_n$, since both sides are linear functions of x . But since $\langle e_i, e_j \rangle = 1$ if $i = j$ and 0 otherwise, this follows from the definition, since

$$\langle e_i, y \rangle = \sum_{j=1}^n f(e_j)\langle e_i, e_j \rangle = f(e_i).$$

Solutions to 3. We need to prove that

$$\lim_{(h,k) \rightarrow 0} \frac{\frac{1+h}{1+k} - 1 - h + k}{\|(h,k)\|} = 0,$$

or equivalently

$$\lim_{(h,k) \rightarrow 0} \left(\frac{1}{1+k} - 1 \right) \cdot \frac{h-k}{\sqrt{h^2+k^2}} = 0.$$

Since the first factor tends to zero, it suffices to prove that the second factor is bounded on $\mathbf{R}^2 \setminus \{(0,0)\}$. Since the numerator equals $\langle (h,k), (1,-1) \rangle$, the Cauchy-Schwarz inequality implies that

$$\left| \frac{h-k}{\sqrt{h^2+k^2}} \right| \leq \sqrt{2}$$

which concludes the proof.

Solutions to 2-7. We claim that f is differentiable at 0 and $Df(0)$ is the constant function equal to zero. For this, we need to prove that

$$\lim_{h \rightarrow 0} \frac{\|f(h) - f(0)\|}{\|h\|} = 0.$$

The assumption implies that $\|f(0)\| \leq \|0\|^2 = 0$, hence $f(0) = 0$. Similarly, $\|f(h)\| \leq \|h\|^2$, and so

$$0 \leq \lim_{h \rightarrow 0} \frac{\|f(h) - f(0)\|}{\|h\|} \leq \lim_{h \rightarrow 0} \|h\| = 0$$

by the squeeze theorem.